Week 4: Array

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Array

Definition

The simplest type of data structure.

Data structure that group a specific type of data into one.

An array is a collection of a fixed number of values of a single type.

Example**:**

For example: if you want to store 100 integers in sequence, you can create an array for it.

int data[100];

Components of Array

ARRAY NAME

SUBSCRIPT/INDEX

DATA TYPE/BASE TYPE

SUBSCRIPT(INDEX)
Common name of variable in array components for element – variable in an array

DIMENSION
The number of subscript required to describe the array

RANGE
The number of element in the array row

LOWER BOUND
The smallest number of the array index type

UPPER BOUND
The highest number of the array index type

ONE DIMENSIONAL ARRAY

$$LOC(i) = BASE + M (I – LB)$$

WHERE:

I = index of an array to be located

BASE = base address

M = memory cell/word per cell

LB = lower boundary of an array

Example 1

What is the location of index 2 in an array OLFU [1…20] with basic address of 220 and the word per cell of 2?

$$LOC(i) = BASE + M (I – LB)$$

$$LOC(2) = 220 + 2 (2-1)$$

$$LOC(2) = 220 + 2(1)$$

$$LOC(2) = 220 + 2$$

$$LOC(2) = 222$$

Example 2

Locate the subscript 20 in an array [10…50] with base address 350 and a memory cell of 5?

$$LOC(i) = BASE + M (I – LB)$$

$$ARRAY(20) = 350 + 5 (20\text{-}10)$$

$$ARRAY(20) = 350 + 5(10)$$

$$ARRAY(20) = 350 + 50$$

$$ARRAY(20) = 400$$

Example 3

Locate index -1 of array [-5…5] with address 375 and memory cell 4.

$$LOC(i) = BASE + M (I – LB)$$

$$ARRAY(-1) = 375 + 4 (-1-(-5))$$

$$ARRAY(-1) = 375 + 4(4)$$

$$ARRAY(-1) = 375 + 16$$

$$ARRAY(-1) = 391$$

Example 4

What is the memory cell of array [3…20] in a base address of 280 and array [5] is located at 286?

$$LOC(i) = BASE + M (I – LB)$$

$$286 = 280 + m (5-3)$$

$$286 = 280 + m(2)$$

$$286 – 280 = m$$

$$6/2 = m$$

$$3 = m$$

More questions?

ASK MS. LEONCIO 

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